

Module 3: Data Sources & Data Connectivity in Power BI

Overview

Data is the foundation of every Power BI report. Before creating visualizations, dashboards, or advanced analytics, you must understand where data comes from and how Power BI connects to different data sources.

This module focuses on data sources, connectivity methods, storage modes, and best practices for importing data into Power BI.

Learning Objectives

After completing this module, you will be able to:

- Understand different types of data sources
 - Connect Power BI to various data sources
 - Understand Import Mode, DirectQuery, and Live Connection
 - Learn data refresh concepts
 - Understand data gateways
 - Connect Excel, CSV, SQL Server, Web APIs, and Cloud Sources
 - Choose the appropriate connectivity mode for business requirements
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1 Introduction to Data Sources

What is a Data Source?

A data source is any location where data is stored and can be accessed for analysis.

Examples:

- Excel Files
 - CSV Files
 - SQL Databases
 - Cloud Applications
 - APIs
 - Websites
-

Why Data Sources Matter

Power BI cannot create reports without data.

The quality, structure, and accessibility of your data source directly impact:

- Report Performance
 - Data Accuracy
 - Dashboard Reliability
 - Business Decisions
-

2 Types of Data Sources

Power BI supports hundreds of data sources.

File-Based Sources

Examples:

- Excel (.xlsx)
 - CSV (.csv)
 - Text Files (.txt)
 - XML
 - JSON
 - PDF
-

Database Sources

Examples:

- SQL Server
 - MySQL
 - PostgreSQL
 - Oracle
 - SQLite
-

Cloud Sources

Examples:

- SharePoint
 - OneDrive
 - Azure SQL Database
 - Google Analytics
 - Salesforce
-

Online Services

Examples:

- Dynamics 365
 - Microsoft Fabric
 - GitHub
 - Adobe Analytics
-

Web Sources

Examples:

- Public APIs
 - Websites
 - JSON Endpoints
-



Power BI Connectivity Architecture

Power BI follows a data flow process:

```
graph TD;
    A[Data Source] --> B[Power BI Connector];
    B --> C[ ];
```

```
Power Query
  ↓
Data Model
  ↓
Visualizations
  ↓
Reports & Dashboards
```

Explanation

Data Source

Stores raw data.

Connector

Allows Power BI to communicate with the source.

Power Query

Performs transformations.

Data Model

Organizes relationships.

Visualization Layer

Displays information through charts and reports.



Storage Modes in Power BI

Power BI provides multiple methods to access data.

Three major storage modes:

1. Import Mode
2. DirectQuery Mode
3. Live Connection

Choosing the correct mode is critical for performance.

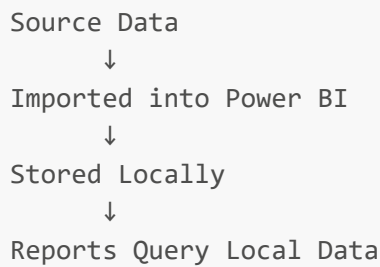


Import Mode

What is Import Mode?

Power BI copies data from the source and stores it inside the Power BI dataset.

How It Works



```
graph TD; A[Source Data] --> B[Imported into Power BI]; B --> C[Stored Locally]; C --> D[Reports Query Local Data];
```

Advantages

Fast Performance

Data is stored inside Power BI.

Advanced Features

Supports:

- DAX
- Calculated Tables
- Calculated Columns

Offline Reporting

Reports work even if the source becomes unavailable.

Disadvantages

Data Becomes Static

Requires refreshes to get new data.

Dataset Size Limitations

Very large datasets may become difficult to manage.

Best Use Cases

- Small to Medium Datasets
 - Historical Reporting
 - High Performance Dashboards
-

6 DirectQuery Mode

What is DirectQuery?

Power BI does not store data locally.

Queries are sent directly to the source whenever users interact with reports.

How It Works

```
graph TD; A[User Clicks Visual] --> B[Power BI Sends Query]; B --> C[Database Processes Query]; C --> D[Results Returned];
```

Advantages

Real-Time Data

Always retrieves current information.

No Large Data Imports

Useful for massive databases.

Disadvantages

Slower Performance

Dependent on database speed.

Limited Features

Some DAX features may not be available.

Best Use Cases

- Real-Time Dashboards
 - Large Enterprise Databases
 - Frequently Changing Data
-

7 Live Connection

What is Live Connection?

Power BI connects directly to an existing analytical model.

Examples:

- SQL Server Analysis Services (SSAS)
 - Azure Analysis Services
 - Semantic Models
-

Characteristics

- No data import
 - No local storage
 - Centralized model management
-

Advantages

Single Source of Truth

All reports use the same model.

Central Governance

Controlled by BI teams.

Disadvantages

Limited Modeling Flexibility

Users cannot modify the model significantly.

8 Comparing Storage Modes

Feature	Import	DirectQuery	Live Connection
Speed	Fast	Medium	Depends
Real-Time Data	No	Yes	Yes
Data Stored in Power BI	Yes	No	No
Supports Full DAX	Yes	Limited	Limited
Best for Small Data	Yes	No	No

Feature	Import	DirectQuery	Live Connection
Best for Large Data	No	Yes	Yes

Connecting to Excel Files

Why Excel?

Excel is one of the most common business data sources.

Steps

Step 1

Home → Get Data

Step 2

Select:

Excel Workbook

Step 3

Choose File

Step 4

Navigator Window Appears

Select:

- Worksheets
- Tables

Step 5

Choose:

Load

or

Transform Data

Best Practice

Convert Excel ranges into Tables before importing.

10 Connecting to CSV Files

What is CSV?

CSV = Comma Separated Values

Example:

```
CustomerID, CustomerName, Sales
101, Rahul, 1000
102, Priya, 1500
```

Steps

Home → Get Data → Text/CSV

Preview Data

Verify:

- Column Names
 - Data Types
 - Delimiters
-

Load Data

Load

or

Transform Data

Connecting to SQL Server

Why SQL Server?

Most enterprise systems store data inside SQL databases.

Connection Steps

Step 1

Home → Get Data → SQL Server

Step 2

Provide:

Server Name
Database Name

Step 3

Choose Connection Mode

- Import
 - DirectQuery
-

Step 4

Authenticate

Methods:

- Windows Authentication
- Database Authentication
- Microsoft Account

Step 5

Load Tables

Example

Sales Table:

OrderID	Product	Revenue
1	Laptop	50000

1 2 Connecting to Web Data

What is Web Data?

Data available through websites or APIs.

Types

HTML Tables

Example:

Tables displayed on websites.

JSON APIs

Example:

```
{  
  "city": "Delhi",  
  "temperature": 42  
}
```

Steps

Home → Get Data → Web

Enter URL

Load or Transform Data.

1 3 Connecting to Cloud Data Sources

Popular Cloud Sources:

- SharePoint
- OneDrive
- Azure SQL Database
- Azure Blob Storage
- Microsoft Fabric

Benefits

Automatic Updates

Data synchronization becomes easier.

Collaboration

Multiple users access the same data.

Scalability

Supports large datasets.

1 4 Data Refresh Concepts

Why Refresh Data?

Source systems continuously change.

Reports must reflect current information.

Manual Refresh

Home → Refresh

Scheduled Refresh

Configured in Power BI Service.

Examples:

- Daily
 - Hourly
 - Weekly
-

Incremental Refresh

Only refreshes new or changed data.

Benefits:

- Faster refreshes
 - Improved performance
-

1 **5** Power BI Gateway

What is Gateway?

A gateway securely connects on-premise data sources to Power BI Service.

Why Needed?

Power BI Service cannot directly access internal company databases.

Gateway acts as a bridge.

Types

Personal Gateway

Single-user scenario.

Standard Gateway

Enterprise environment.

Multiple users supported.

1 **6** Best Practices

Choose Correct Storage Mode

- Small Data → Import
- Large Data → DirectQuery
- Enterprise Model → Live Connection

Validate Data Types

- Always verify:
- Numbers
 - Dates
 - Currency Fields

Use Power Query

Clean data before loading.

Remove Unnecessary Columns

Improves performance.

Use Incremental Refresh

For large datasets.

Key Terminologies

Term	Definition
Data Source	Location where data is stored
Connector	Tool used to connect to data
Dataset	Collection of imported data
Import Mode	Data stored inside Power BI
DirectQuery	Queries sent directly to source
Live Connection	Connection to external model
Refresh	Updating dataset with latest data
Gateway	Secure bridge to on-premise data
API	Application Programming Interface

Term	Definition
Cloud Source	Data stored in cloud platforms



Practice Exercises

Exercise 1

List five data sources commonly used in businesses.

Exercise 2

Compare:

- Import Mode
 - DirectQuery
 - Live Connection
-

Exercise 3

Connect an Excel file to Power BI.

Exercise 4

Connect a CSV file and verify data types.

Exercise 5

Research Power BI Gateway and explain its purpose.



Mini Project

Multi-Source Data Import

Objective

Import data from multiple sources.

Tasks

Import:

- Excel File
- CSV File

Verify:

- Column Names
- Data Types
- Data Quality

Create:

- One Table Visual
- One Bar Chart



Module Summary

In this module, you learned:

- ✓ Types of Data Sources
- ✓ Power BI Connectivity Architecture
- ✓ Import Mode
- ✓ DirectQuery Mode
- ✓ Live Connection
- ✓ Connecting Excel Files
- ✓ Connecting CSV Files
- ✓ Connecting SQL Server
- ✓ Connecting Web Data
- ✓ Cloud Data Sources
- ✓ Data Refresh Concepts
- ✓ Power BI Gateway

These concepts form the foundation for data preparation, which will be covered in the next module using Power Query.



Next Module

Module 4: Power Query Editor & Data Transformation

Topics Covered:

- Introduction to Power Query
- Data Cleaning
- Data Transformation

- Query Folding
- Data Profiling
- Merge Queries
- Append Queries
- Pivot & Unpivot Operations
- Best Practices for Data Preparation